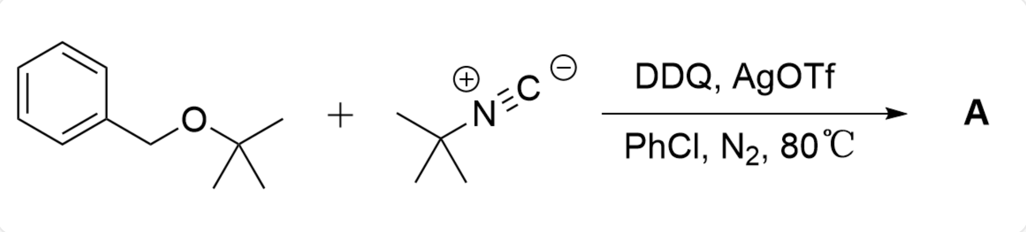
Question

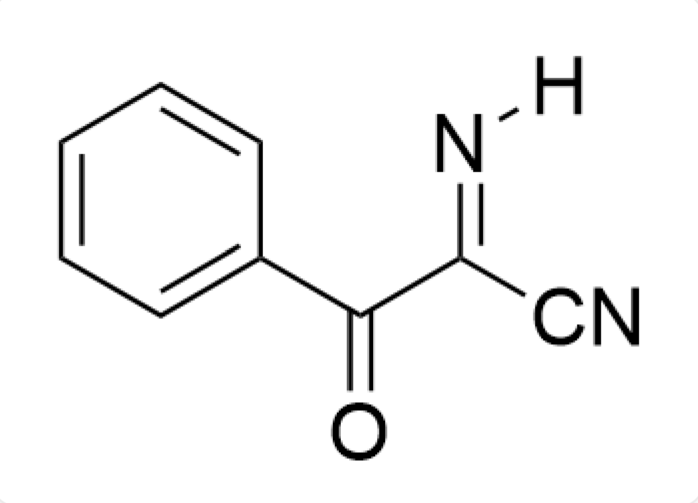


CC(C)(C)OCC1=CC=CC=C1.CC(C)([N+]#[C-])C>[DDQ],[AgOTf],[PhCl],[N_2]>[A], A is the reaction product, the reaction temperature is $80^{\circ}\mathrm{C}$

Where the amount of DDQ and tert-butyl isocyanide is at least twice the amount of benzyl tert-butyl ether. Mechanistic studies show that the reaction is not a free radical reaction mechanism. Deduce the structural formula of the reaction product **A** (without considering stereoisomers).

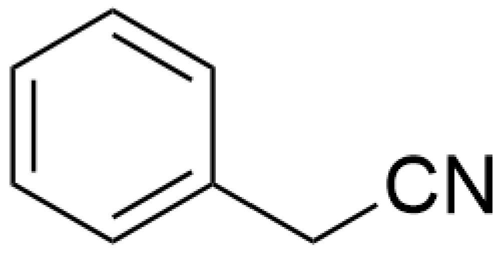
A. All other options are incorrect

B.



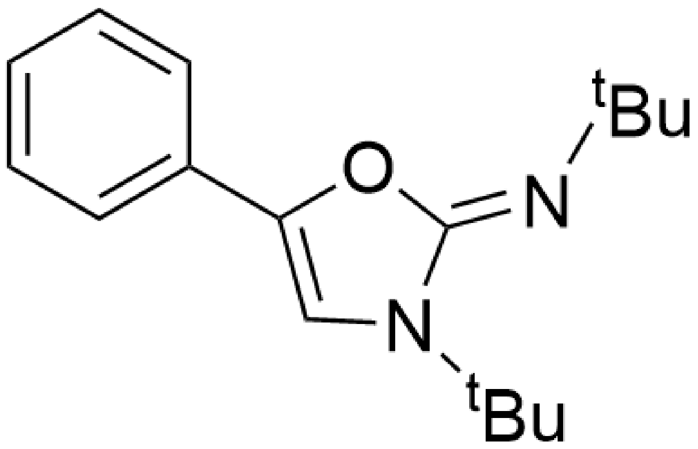
[H]/N=C(C#N)/C(C1=CC=CC=C1)=O

C.



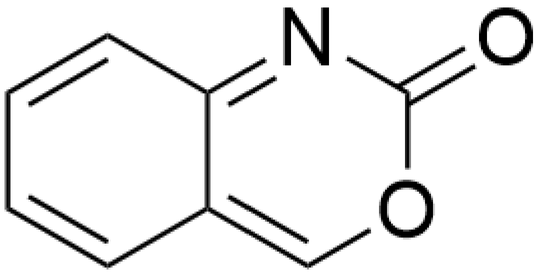
N#CCC1=CC=CC=C1

D.



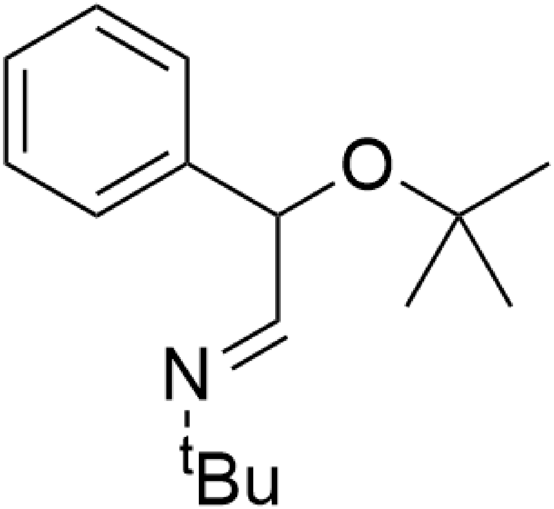
CC(/N=C1N(C(C)(C)C)C=C(O/1)C2=CC=CC=C2)(C)C

E.



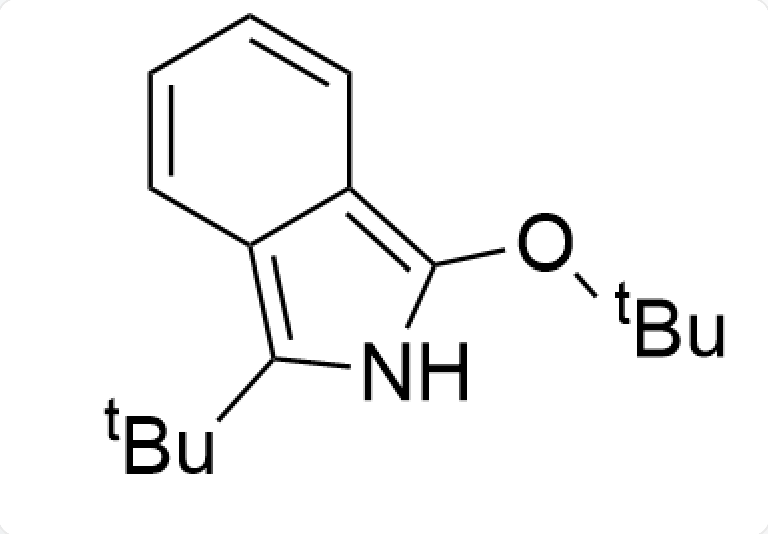
O=C1OC=C2C=CC=CC2=N1

F.



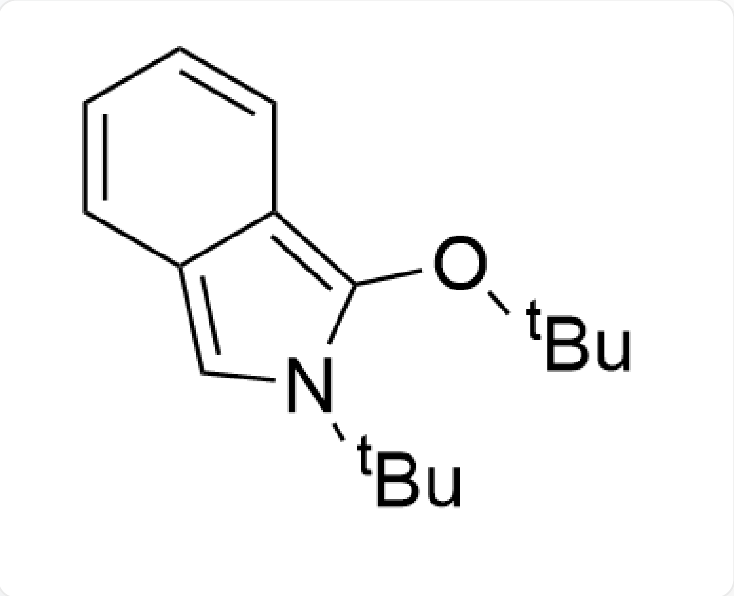
CC(C)(C)OC(/C=N/C(C)(C)C)C1=CC=CC=C1

G.



CC(OC1=C2C(C=CC=C2)=C(C(C)(C)C)N1)(C)C

H.



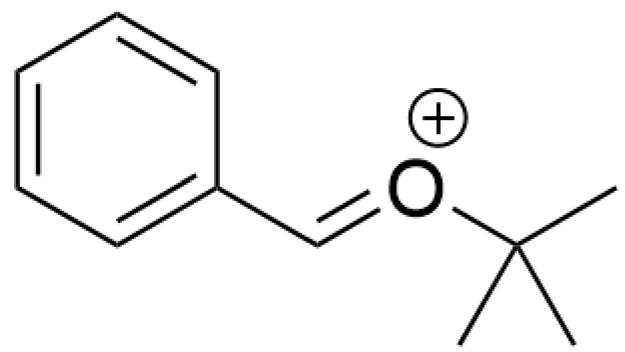
CC(OC1=C2C(C=CC=C2)=CN1C(C)(C)C)(C)C

Answer

Correct Answer: A

Detailed Explanation

First, the benzyl ether substrate is oxidized by DDQ to form intermediate 1

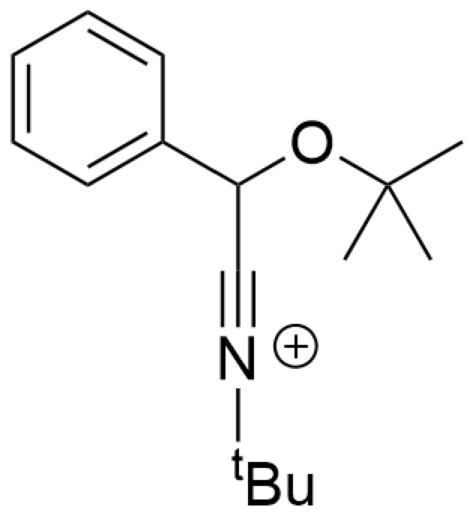


CC(C)(C)/[O+]=C/C1=CC=CC=C1

CHECKPOINT 1 PTS

CC(C)(C)/[O+]=C/C1=CC=CC=C1

Subsequently, a one-step addition reaction occurs to obtain intermediate 2

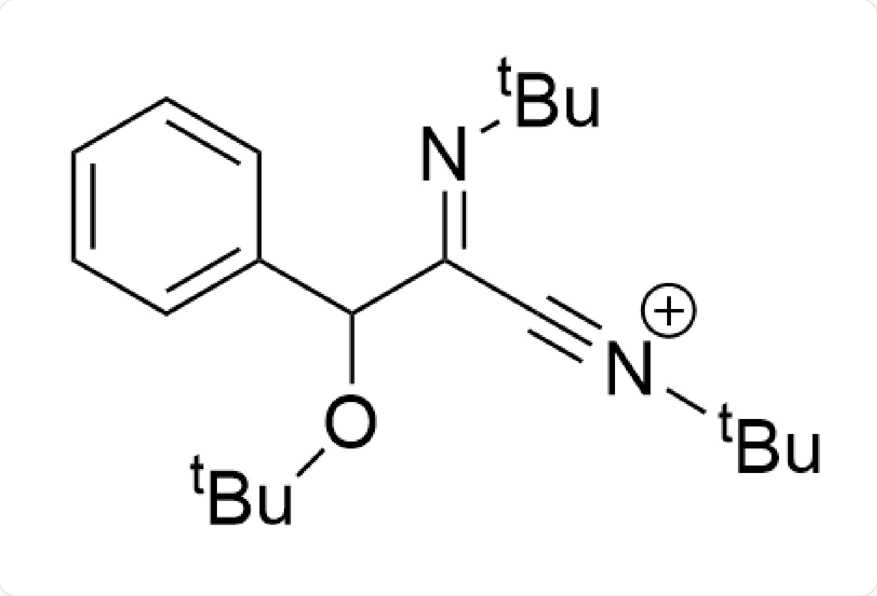


CC(C)(C)OC(C#[N+]C(C)(C)C)C1=CC=CC=C1

CHECKPOINT 1 PTS

CC(C)(C)OC(C#[N+]C(C)(C)C)C1=CC=CC=C1

A molecule of addition reaction occurs again to obtain intermediate 3

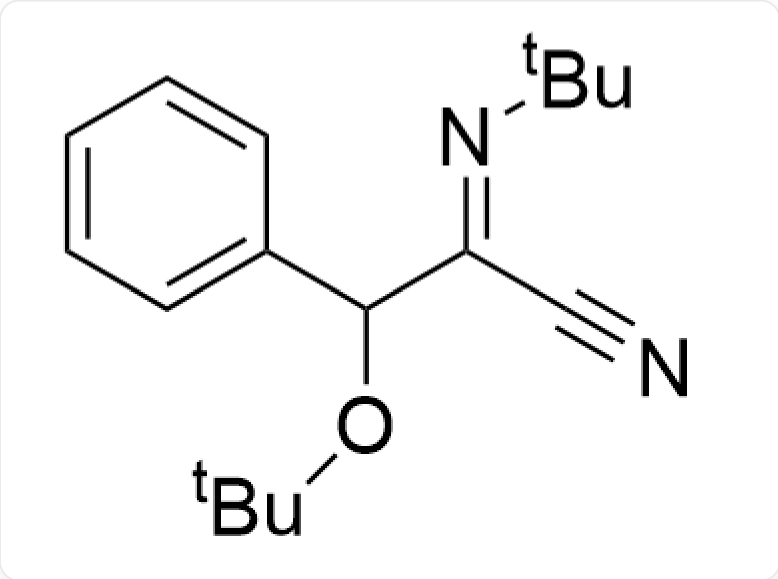


CC(OC(/C(C#[N+]C(C)(C)C)=N/C(C)(C)C)C1=CC=CC=C1)(C)C

CHECKPOINT 1 PTS

CC(OC(/C(C#[N+]C(C)(C)C)=N/C(C)(C)C)C1=CC=CC=C1)(C)C

Subsequently, a tert-butyl carbenium ion is eliminated to obtain intermediate 4

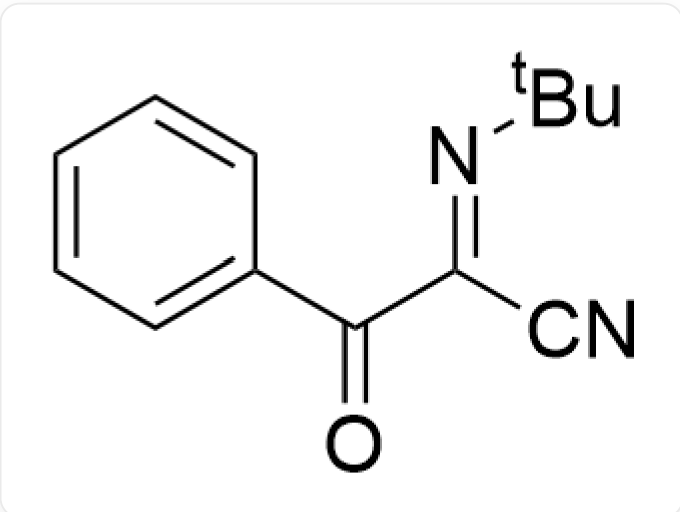


N#C/C(C(OC(C)(C)C)C1=CC=CC=C1)=N\C(C)(C)C

CHECKPOINT 1 PTS

N#C/C(C(OC(C)(C)C)C1=CC=CC=C1)=N\C(C)(C)C

Finally, it is oxidized again by a molecule of DDQ to obtain the reaction product **A**



O=C(/C(C#N)=N/C(C)(C)C)C1=CC=CC=C1

CHECKPOINT 1 PTS

O=C(/C(C#N)=N/C(C)(C)C)C1=CC=CC=C1